

A Robust and Explainable CNN Framework for Automated MRI Distortion Detection with Grad-CAM Interpretation

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ABSTRACT

Magnetic Resonance Imaging (MRI) is an essential technique for present medical diagnosis, due to its high-resolution internal body imaging capability. Nevertheless, image quality of an MRI image could be deteriorated by distortions, such as noise, motion blur, which hinder effective automatic image analysis, thus leads to unreliable detection and diagnosis. In this paper, we present a Convolutional Neural Network (CNN) based framework for MRI image distortion detection. To imitate practical condition, artificial distortions, namely Gaussian noise and Gaussian blur are added to the original images to let the model learning features that discriminate between good and degraded images. The designed system achieved high classification accuracy with high confidence estimation. In addition to high performance, Grad-CAM is introduced for model explanation and to improve the interpretability for medical diagnosis. The generated activation maps can show most relevant regions for the decision making, which provides good model interpretability. Overall, our proposed approach serves as an effective pre-processing step for medical image processing pipeline.

Keywords: CNN, MRI, Image Distortion, Noise, Blur, Deep Learning, Medical Imaging

1. INTRODUCTION

Medical imaging has developed to become a significant pillar in contemporary medicine by providing tools to assess and manage a wide range of pathologies effectively. MRI, as a key medical imaging technique that can achieve high-resolution images of soft tissues with non-ionizing radiation, provides a significant resource for the assessment and diagnosis of neurological

diseases, tumours and structural lesions in the body, although subject to a number of artifacts in the imaging process. Image distortions could be caused by patient movement, hardware constraints, external interference or transmission errors. The commonly encountered distortion types are Gaussian noise, motion blur and compression artifacts, all of which can impair image quality, and negatively affect the accuracy and effectiveness of computer-aided medical image analysis systems. In particular, the diagnostic model using deep learning techniques requires the input to be of highest quality, and would be less precise in accuracy if fed distorted images.

In recent years, deep learning approaches have achieved significant success in many medical image processing applications including classification, segmentation and detection. Among all deep learning architectures, Convolutional Neural Networks (CNNs) have become particularly popular. CNNs can automatically learn hierarchical attributes from raw medical images, which is efficient to extract high-level features. Previous researches successfully adopted CNNs for disease detection like breast cancer and brain tumour using MRI.

Currently most of the work in this field is concerned with disease detection rather than image quality estimation. However, quality of the input images is a critical aspect that directly determines the accuracy of diagnostic systems. Even a correct model may generate a wrong diagnostic, if input image quality is affected. This means that there must be some mechanism to check the quality of MRI images before these images are used for analysis.

A further significant limitation of deep learning models in medical usage is their non-interpretability. Black box models, in a sense,

make it difficult for a clinician to understand why the prediction was made. To mitigate this problem, explainability methods, such as Grad-CAM were developed. Grad-CAM produces visual heatmaps in which crucial areas of an image that affecting the model's decision will highlight, thereby improving transparency and trust.

In this paper we provide a CNN-based framework for distortion detection in MRIs along with explainability via Grad-CAM. Our system classifies images as "good" or "distorted" and produce explanations as visual images. Such approach enhances the accuracy of automated medical imaging system while keeping it as understandable, and therefore as clinically relevant as possible.

2.LITERATURE REVIEW

Recent studies have increasingly focused on improving distortion correction and preprocessing reliability in the neuroimaging data. In a study by Feng et al. [1] provide a rapid and robust framework for correcting the susceptibility-induced distortion especially when missing modality is involved. Li et al. [2] in another studies, explained how the variance from preprocessing could impact the predictive model for fMRI, and indicated that distortion correction is important especially in functional connectivity-based predictions. Before these recent works, Viard et al. [3] investigated geometric distortion correction for low-field MRI and Zarrabi et al. [4] proposed distortion compensation in wavelet domain for the safe and robust processing of medical image. Getting further into deep learning-based reconstruction techniques, Shan et al. [5] developed a deep unrolled network to held account for non-homogeneous MRI reconstruction.

For multimodal data fusion and image registration an unsupervised framework was introduced by Zhong [6] aims to fuse misaligned PAT and MRI images with cross-modality generation and registration for the spatial consistency. Also, Yogeshkumar [7] in his research, developed a deformable convolution networks for the reconstruction of misaligned MRI data for achieving exact brain tumor classification. An alignment-guided forward-distortion model was proposed by Kayapinar et al [8] to deal with susceptibility artifact correction in EPI images and it showed the capability of unsupervised deep learning methods in medical image processing. GAN based approach was also used by Vashisth et al [9] to generate synthetic

brain MRI images using data augmentation and robust application in medical imaging.

Additionally, the introduction of explainable AI techniques to the multimodal learning approaches have gained significant interest recently for improving the trust and diagnostic capability of the models. Basheer et al. [10] introduced a Grad-CAM-based explainability approach incorporated within an ensemble model to perform brain MRI analysis. Similarly, Ketabi et al. [11] have explored the advantages of supervised multimodal learning in diagnosing brain tumors in children, indicating performance improvements and explainability with combined representations of information. Lastly, Upadhyay et al. [12] introduced a framework that aims to establish transparency and trust in an AI model using brain MRIs, also re-iterating the crucial role of explainable AI. All these works together mark a path from correction, distortion, and reconstruction to multimodal fusion and intelligence.

3.PROPOSED METHODOLOGY

Our designed methodology contains an organized pipeline as follows: dataset preparation, preprocessing, synthesis of artificial distortions, feature extraction and classification by CNN, and explanation by Grad-CAM. The overall framework ensures accurate detection of distorted MRI images while providing interpretable results an figure 1 represent it.

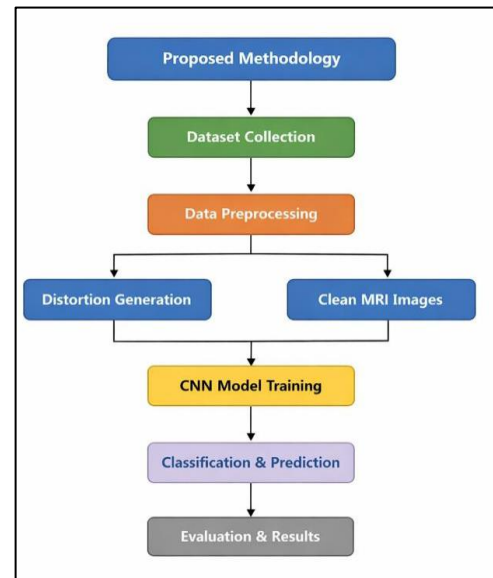


Figure 1: Overall workflow of the proposed model

3.1 Dataset Preparation

MRI pictures from publicly accessible sources make up the dataset used in this investigation. The base dataset is made up of these high-quality scan images. Artificial distortions are introduced to simulate real-world scenarios because there aren't many real-world distorted datasets.

There are two classes in the dataset:

- Good Images: Unaltered original MRI images
- Images with artificially added distortions are referred to as distorted images.

The model can learn distinguishing characteristics between clear and degraded images thanks to this classification.

3.2 Data Preprocessing

Pixel normalization is performed using min-max scaling to ensure numerical stability during training carried out to guarantee consistency and enhance model performance prior to feeding the images into the model. These actions consist of:

- Resizing: Resize all images to a fixed size (e.g. 224X224) such that they match the size required as input to the CNN.
- Normalization: Rescale the pixel values so they are within 0-1. This helps stabilize the training.
- Data Augmentation: Methods such as rotation and horizontal flipping can be performed to increase the variety and decrease the amount of over fitting.

These preprocessing procedures aid in enhancing the model's capacity for generalization.

3.3 Generation of Distortion

Artificial noise and blur are applied to the original MRI images to replicate real-world distortions. The intensity of noise and blur is varied using different standard deviation values to simulate multiple real-world distortion levels. The following methods are applied:

- Gaussian Noise: To mimic sensor or environmental noise, random noise is added to the image pixels.
- Gaussian Blur: To mimic motion blur or focus problems, a smoothing filter is used.

The model is able to learn robust features because these distortions are applied at different intensities to create a diverse dataset.

3.4 CNN Model Architecture

In the case of classification, the proposed system of figure 2 employs a Convolutional Neural Network (CNN). CNNs are particularly adept at spatial hierarchies and are a logical choice for image tasks. The CNN consists of layers of 3x3 kernels with ReLU and max-pooling for hierarchical feature extraction.

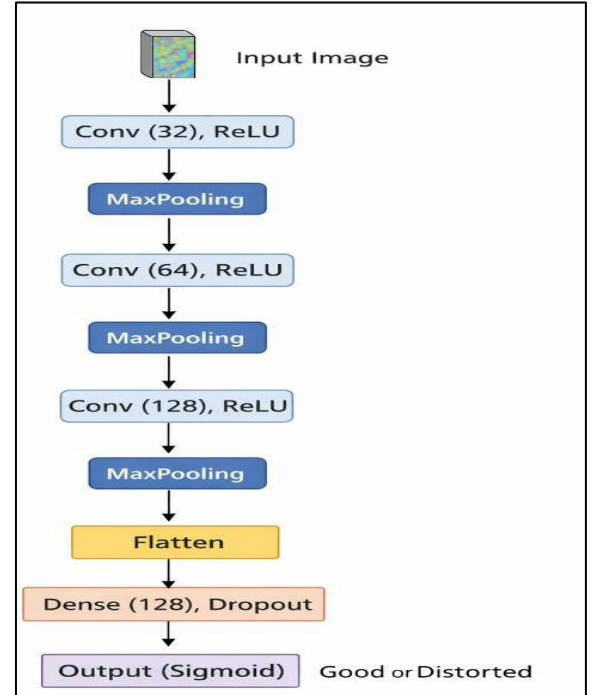


Figure 2: CNN Architecture Model

The following layers make up the architecture:

- Convolutional Layers: Extract characteristics from the input images, such as edges, textures and patterns. In order to express the operations in the CNN in a mathematical way, fundamental equations related to feature extraction, activation, and pooling are presented and equation(1) describe how the model processes MRI images and learns relevant distortion patterns. And equation 1 formulate it

$$F(x, y) = \sum I(x, y).K(x, y) \quad (1)$$

Where $I(x, y)$ represents the input MRI image, $K(x, y)$ represents the convolution kernel, and $F(x, y)$ is the extracted feature map.

- By introducing non-linearity, the activation function (ReLU) enables the network to learn intricate relationships and mathematically it is represented as equation 2,

$$f(x) = \max(0, x) \quad (2)$$

Where, x is the input and $f(x)$ is the output after activation.

- Pooling Layers (Max Pooling): Lower computational complexity by reducing the spatial dimensions of feature maps and represented as equation(3)

$$P = \max(F) \quad (3)$$

- Fully Connected Layers: Use extracted features to perform classification.
- Output Layer: Classifies images as "good" or "distorted" using a sigmoid or softmax activation function.

3.5 Model Training

For the CNN model training, we used labelled data. During the training process:

- Training and validation sets are separated from the dataset.
- The model is learned by minimizing a loss function like binary cross-entropy. At each step of the learning phase the parameters are updated so that the accuracy of the model on a loss function, defined as the difference between the actual and predicted outputs (4), is maximized.

$$L = -[y \log(p) + (1 - y) \log(1 + p)] \quad (4)$$

Where y is the true label and p is the predicted probability.

- Model weights are updated using an optimizer like Adam.
- Training is carried out over several epochs.
- To assess model performance during training, performance metrics like accuracy and loss are tracked.

3.6 Forecasting and Assessment

Model evaluation is done based on untested data following training. The system offers: Predicted Class: Distorted or Good Confidence Score, Prediction Probability values Precision Recall Accuracy. As a preprocessing step to eliminate low-quality images, this system can be incorporated into medical imaging workflows.

3.6. Explainability Using Grad-CAM

Grad-weighted Class Activation Mapping (Grad-CAM) is added into the framework for

interpretability. Grad-CAM produces heat maps which localize areas of the MRI images that have the largest influence on the model's prediction and mathematical expression is equation (5)

$$L = ReLU \sum_k A^k \quad (5)$$

The working process: The gradients of the scores of target class are computed against feature maps of the last convolutional layer. Importance weights are allocated for each feature map based on gradients and these weighted feature maps are used to generate a localization map. The resultant heatmap is overlaid over the original MRI image. The whole explainability and localisation process using Grad-CAM is shown in figure(3).

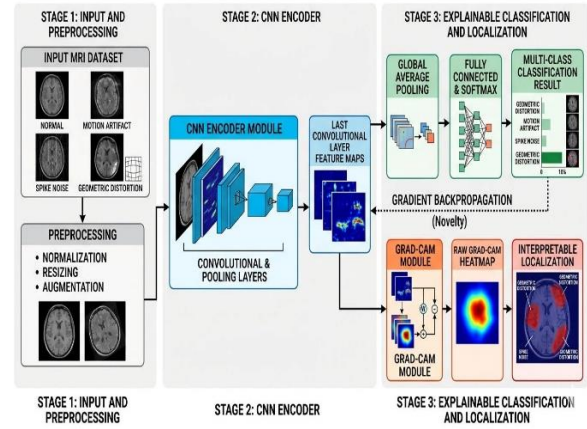


Figure 3: Localization Stages using Grad-CAM

This visualization helps verify whether the model focuses on distortion-relevant regions such as noise patterns or blurred structures, thereby improving transparency and reliability in medical image analysis.

4. EXPERIMENTAL SETUP AND DATA DISTRIBUTION

In this section, describe the experimental setup adopted for training and evaluating the CNN-based MRI distortion detection model and used parameters mentioned in Table 1. Beyond the classification accuracy, the interpretability is implemented through Grad-CAM that provides visual interpretation of the decisions. A useful component we added to the model was Grad-CAM to improve the model interpretability, by showing which areas have influenced the result the most. This is very important in the field of medical imaging, so that the user can have confidence in the results produced. Through experiments we have proven that both a high performance and interpretability can be obtained.

Table 1: Experimental Configuration

Image Type	Highlighted Regions	Interpretation
Good Image	Uniform regions	No distortion detected
Noisy Image	Scattered pixel regions	Noise correctly identified
Blurred Image	Edge-deficient regions	Blur regions highlighted
Mixed Distortion	Multiple affected areas	Complex distortions detected

5.RESULT AND DISCUSSION

5.1 Training Performance Analysis

The convergence of CNNs was checked monitoring how the performance of training and validation. Overfitting was avoided by early stopping and validation monitoring. Accuracy and loss were used as indicators for learning speed and generalization as shown in figure 4.

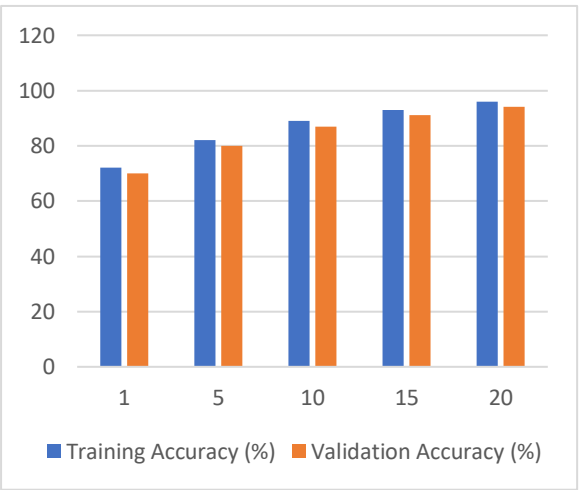


Figure 4: Training & Validation Accuracy

The model improved constantly with epochs. The match of training and validation accuracy show that the generalization is good, which means that the model has learned visual features and it has not just memorized the distortion.

5.2 Evaluation Metrics

The accuracy of the model (94.5%) reflects its overall performance and appears good and reference mentioned in Figure 5. High recall (95.2%) in the medical imaging context is an essential characteristic as it indicates most distorted images have been identified. Good precision and F1 score show that the model performs consistently with no preference to a particular class. Both accuracy and recall show the model consistently identifies distorted images correctly and a high F1 score means the balance between precision and recall is retained.

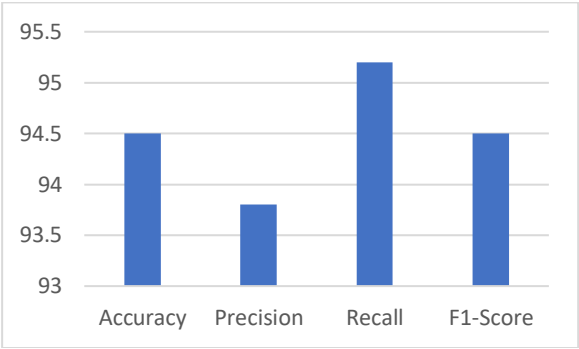


Figure 5: Performance Metrics

5.3 Confusion Matrix Analysis

The low number of errors show that the model is stable. Even less images classified as good are actually distorted, which decrease the possibility of misdiagnosis during subsequent analysis of medical data and is represented in Table 2.

Table 2: Confusion Matrix

X	Predicted Good	Predicted Distorted
Actual Good	180	10
Actual Distorted	8	202

5.4 Performance Across Distortion Types

Robustness assessment: the model is tested on 3 kinds of distortions: Gaussian noise, blur, and mixed distortions. Performance on Gaussian noise is slightly better than the other two, because the distinctive differences between pixels are well captured by CNNs. The performance degradation on blur means smooth distortions are difficult. Mixed distortion raises complexity and therefore, slightly decreases the accuracy and figure 6 represents it.

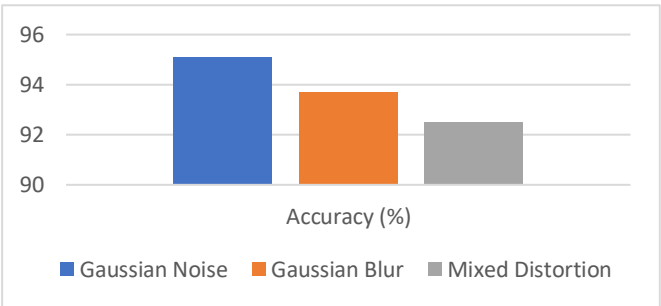


Figure 6: Accuracy per Distortion Type

5.5 Explainability Analysis Using Grad-CAM

The Grad-CAM method is applied so as to know which regions are the most relevant for the prediction by the model from the MRI image. The model is thus able to find if it is focusing on the

damaged areas. The overall summary mentioned in Table 3.

Table 3: Grad-CAM Interpretation Summary

Parameter	Value
Image Size	224 × 224
Batch Size	32
Epochs	20–30
Optimizer	Adam
Learning Rate	0.001
Loss Function	Binary Cross-Entropy
Train-Test Split	80% – 20%
Explainability Tool	Grad-CAM

Grad-CAM images show the model focuses on significant regions of distortion, not unrelated background areas. Noisy images, activation maps, show scatter at the pixel level, indicative of the noise present in the image. Blurry images highlight areas where edges become less clear.

This indicates that the CNN is not operating as a "black box" but relies on reasonable and semantic attributes of the image. Such an interpretation system is necessary in medical systems so that users of medical devices would have confidence in automated systems.

To enhance interpretability, Grad-CAM is applied to visualize the relevant regions of an MRI image which influence the prediction. This allows verification of whether the model focuses on distortion-related regions.

5.6 Comparative Analysis with Existing Methods

The proposed model is compared with traditional machine learning approaches to validate its effectiveness and it shown in figure 7.

The CNN is superior to traditional approaches due to automatic spatial hierarchy learning. To improve on this already robust CNN model we integrate the visualization of where the model is concentrating its processing using Grad-CAM which traditional approaches lack.

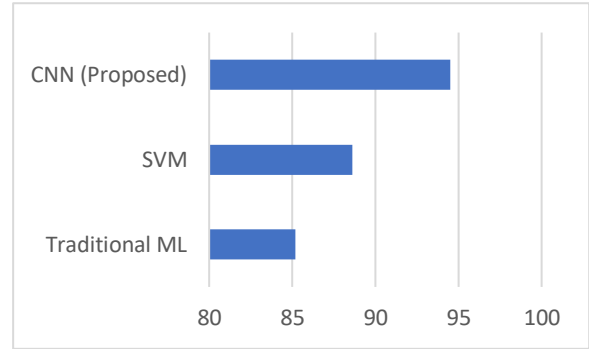


Figure 7: Model Comparison

6. CONCLUSION AND FUTURE SCOPE

The objective of this work was to implement a framework based on deep learning, and in particular on Convolutional Neural Networks (CNN), in order to perform Magnetic Resonance Imaging (MRI) distortion detection. The implemented CNN can classify images between good-quality ones and distorted ones, and obtain optimal results in terms of precision and well-balanced results for evaluation indexes. Enhancing with Explainability. Adding Grad-CAM would allow another interesting level by displaying decision of the model.

The created heat maps indicate that the model learns to pay attention to areas relevant to distortion and therefore improve transparency and trust for medical applications. It can be used as a pre-processing step in an automated diagnosis system and can be used to feed only quality MR images to downstream classifiers.

In terms of future work, we aim to enrich the robustness by using actual clinical images with variety of distortion types to train the model. Further work can also generalize the framework to multiple types and levels of distortions. By utilizing advanced neural network architecture such as attention-based models and hybrid networks, higher performance can be reached. Integrating this system into end-to-end medical imaging pipeline, with real time deployment on edge, would increase the utility in a real medical setting.

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